Capital University

Faculty of Engineering

Computer and system Department

Smart Meter Monitoring Mobile Application

Course Name: Mobile Computing

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Introduction

Project Overview

In the modern era of rapid urbanization and the global energy crisis, efficient power management is no longer a luxury but a necessity. Traditional energy meters provide users with a "single total value" at the end of the month, leaving homeowners in the dark about their actual consumption patterns. The **Smart Meter Monitoring System** is an advanced IoT and AI-driven mobile platform designed to bridge this information gap, transforming raw electrical data into actionable intelligence.

The Problem Statement

The primary challenge in home energy management is the lack of "Appliance-Level Transparency." Most users are unaware that a malfunctioning air conditioner or a forgotten water heater can account for over 40% of their monthly bill. Current systems fail to:

1. **Disaggregate loads**: They cannot tell the difference between a refrigerator and a washing machine.
2. **Detect Anomalies**: They do not alert users to dangerous power surges or "leaks" in real-time.
3. **Predict Future Costs**: Users have no way to forecast their bill before it arrives.

The Proposed Solution

Our project introduces a high-performance **React Native** application integrated with a **Non-Intrusive Load Monitoring (NILM)** AI engine. By analyzing the high-frequency voltage and current signatures from a single entry point, our system can identify individual appliances without requiring expensive sensors on every plug.

Key Innovations Included:

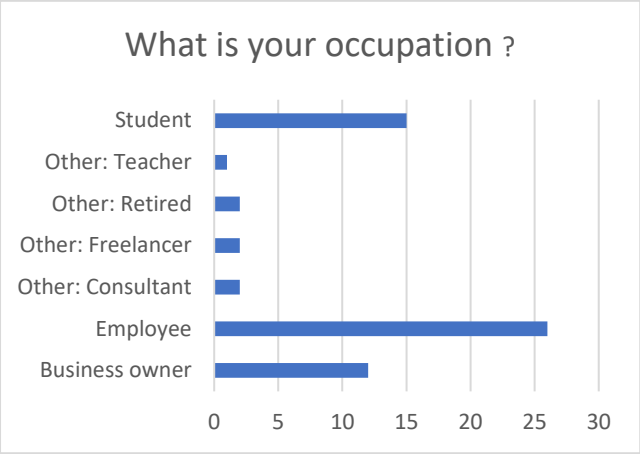
- **AI Disaggregation (NILM)**: Using deep learning models hosted on Hugging Face to "see" which appliances are ON.
- **Real-Time Anomaly Detection**: An intelligent monitoring system that triggers immediate "Rose-Alert" notifications when unusual energy spikes are detected.
- **Dynamic Energy Forecasting**: A predictive engine that analyzes historical patterns to give users an accurate kWh forecast for the upcoming day, week, or month.
- **Glassmorphic Design Interface**: A premium, motion-staggered UI that makes complex energy data simple, intuitive, and engaging for the everyday user.

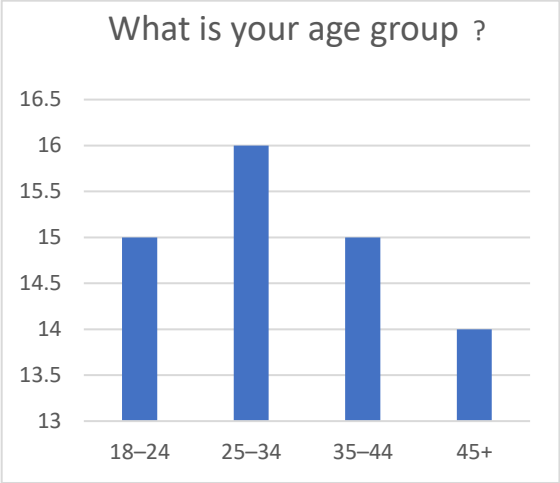
Impact and Sustainability

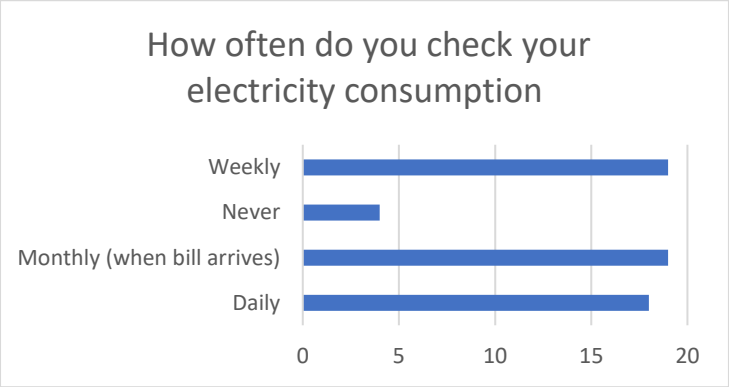
By providing real-time transparency, this system empowers users to reduce their carbon footprint and lower their monthly expenses by up to 20-30%. It transitions the user from a "Passive Consumer" to an "Active Energy Manager," contributing directly to the global goal of sustainable living and smart city development.

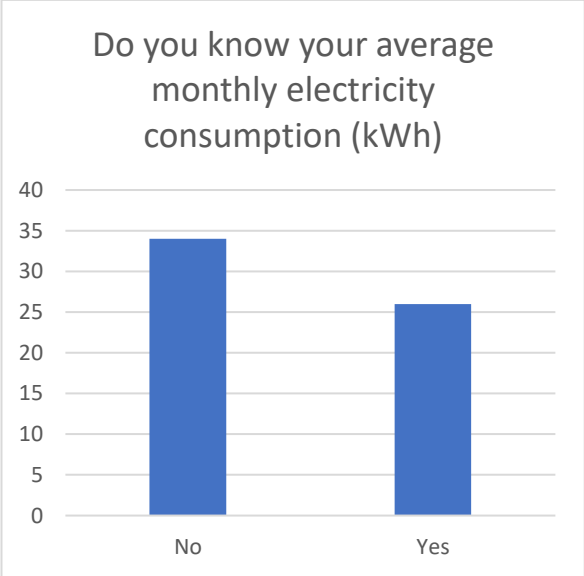
Stage 01:UX Phase (Subsections: UX stage I, II, III, and IV)

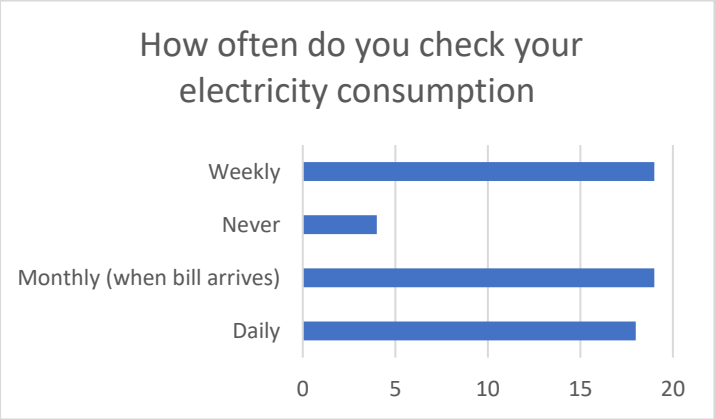
Stage I: Research

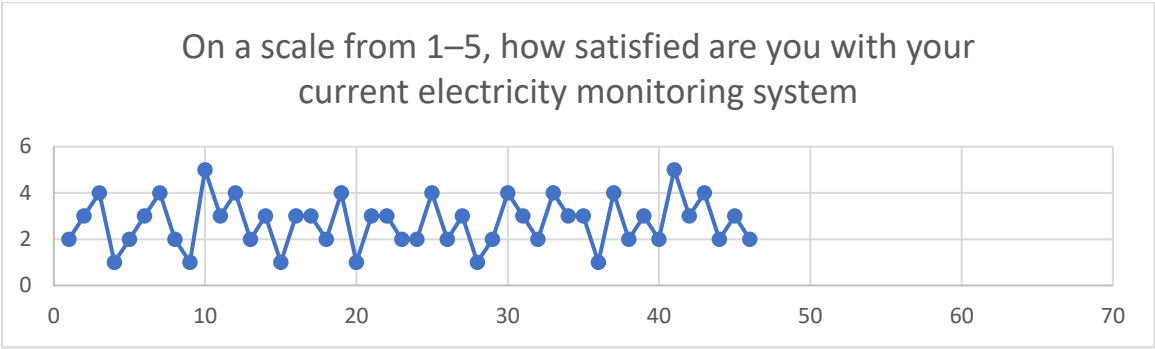


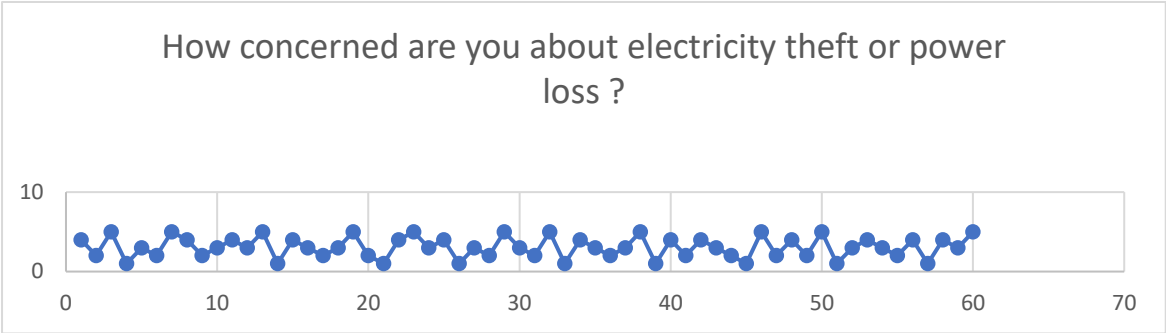


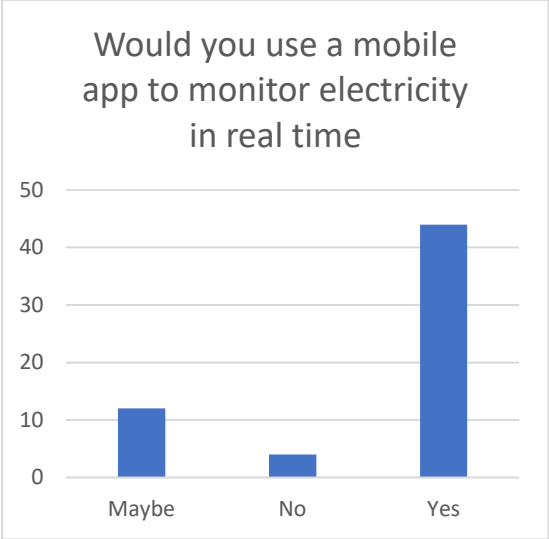


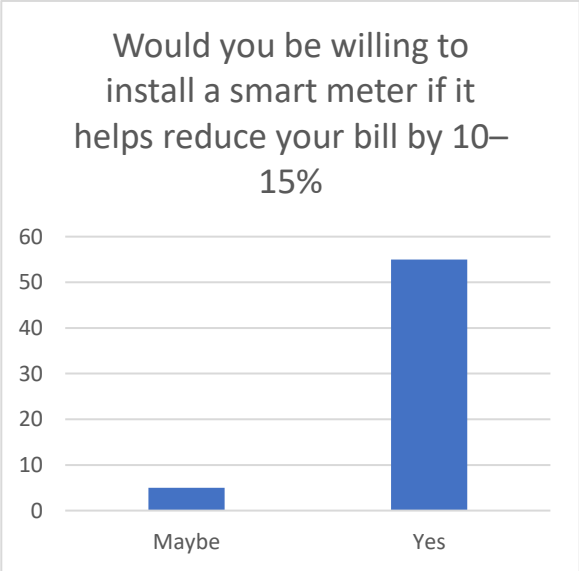


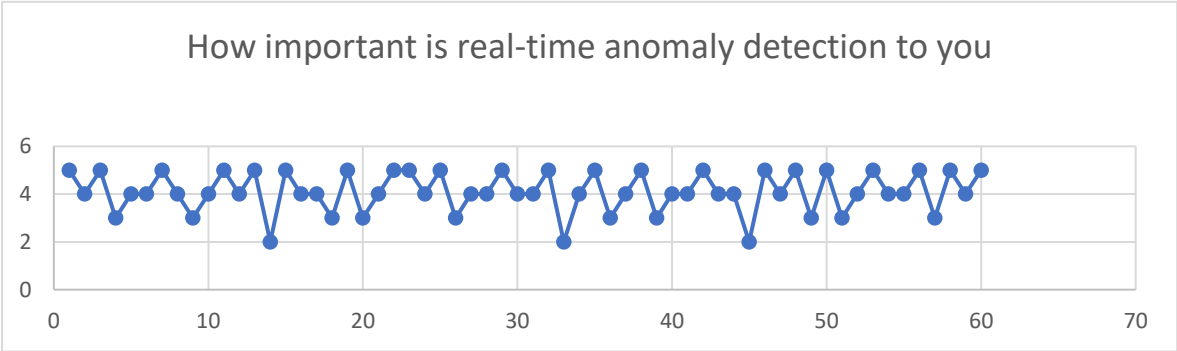



On a scale from 1–5, how satisfied are you with your current electricity monitoring system
6
4
2
0
0
10
20
30
40
50
60
70


How concerned are you about electricity theft or power loss ?
10
0
0
10
20
30
40
50
60
70


Would you use a mobile app to monitor electricity in real time
50
40
30
20
10
0
Maybe
No
Yes


Would you be willing to install a smart meter if it helps reduce your bill by 10–15%
60
50
40
30
20
10
0
Maybe
Yes


How important is real-time anomaly detection to you
6
4
2
0
0
10
20
30
40
50
60
70

Empathy Map — Ahmed Hassan

The empathy map consolidates research insights into four quadrants — what Ahmed Says, Thinks, Does, and Feels — providing a holistic understanding of his emotional and behavioral relationship with energy management.

Empathy Map: Ahmed Hassan

Says

- I never know my electricity consumption until the bill arrives
- Sometimes my bill is much higher than I expect
- I wish I could see which devices use the most electricity
- Current electricity monitoring systems are confusing or hard to use
- It's hard to track my energy usage on a daily basis
- I want alerts if something abnormal happens with my electricity
- I don't understand my bill calculations
- I wish there was an easy way to save money on electricity

Thinks

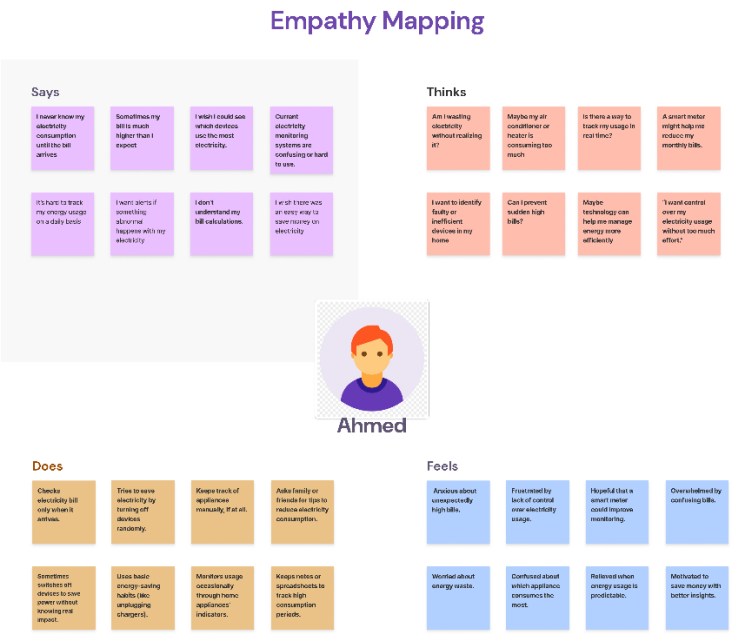
- Am I wasting electricity without realizing it?
- Maybe my air conditioner or heater is consuming too much
- Is there a way to track my usage in real time?
- A smart meter might help me reduce my monthly bills
- I want to identify faulty or inefficient devices in my home
- Can I prevent sudden high bills?
- Maybe technology can help me manage energy more efficiently
- I want control over my electricity usage without too much effort

Does

- Checks electricity bill only when it arrives
- Tries to save electricity by turning off devices randomly
- Keeps track of appliances manually, if at all
- Asks family or friends for tips to reduce electricity consumption
- Sometimes switches off devices to save power without knowing real impact
- Uses basic energy-saving habits (like unplugging chargers)
- Monitors usage occasionally through home appliances' indicators
- Keeps notes or spreadsheets to track high consumption periods

Feels

- Anxious about unexpectedly high bills
- Frustrated by lack of control over electricity usage
- Hopeful that a smart meter could improve monitoring
- Overwhelmed by confusing bills
- Worried about energy waste
- Confused about which appliance consumes the most
- Relieved when energy usage is predictable
- Motivated to save money with better insights



Persona

The following persona was developed through user research to represent the primary target user of the Smart Meter Mobile Application. Ahmed's profile informs all subsequent design and problem-definition decisions.

Ahmed Hassan

Name: Ahmed Hassan
Age: 35
Education: Masters
Occupation: Homeowner
Location: Cairo, Egypt
Status: Married

Bio

Ahmed is a tech-savvy homeowner who wants better control and visibility over his household energy consumption. He is concerned about energy waste, cost, and ensuring his family's comfort.

Interests

- Smart home technology
- Energy efficiency
- IoT devices
- Renewable energy

Goals

- Monitor household energy consumption in real-time
- Reduce electricity bills
- Ensure devices operate efficiently
- Detect unusual consumption

Needs & Expectations

- Reliable, real-time monitoring

Pain Points

- Unreliable smart meters

– Alerts for unusual usage – Secure and easy-to-use interface (mobile & web) – Integration with other smart home devices	– Lack of transparency in energy consumption – Fear of cyber threats on connected devices – Difficulty detecting appliance faults

Motivation	Challenges
– Save money on energy bills – Protect his home and appliances – Contribute to sustainable energy use	– Understanding complex energy data – Dealing with unreliable or inaccurate IoT sensors – Limited control over energy distribution in older homes

Stage 02: Define a Problem

Stage 02 of the Design Thinking process focuses on synthesizing research findings into a clear, actionable problem statement. By defining pain points, answering the 5Ws and H, and framing the problem from the user's perspective, we ensure that subsequent design solutions address genuine user needs.

3.1 Pain Points & Categories

Pain points are any UX issues or friction that frustrate the user and block them from getting what they need. Based on Ahmed's persona and empathy map, the following pain points were identified and categorized:

Ahmed's Key Pain Points
• Overwhelming lack of real-time visibility into daily electricity consumption • Unexpectedly high bills with no clear explanation of what caused the spike • Unreliable or confusing existing monitoring systems and smart meters • No automatic alerts when consumption is unusually high or abnormal • Limited understanding of which appliances are the biggest energy consumers • Fear of cyber threats and data privacy concerns with connected IoT devices

Pain points fall into four recognized UX categories:

Category	Description
Financial	Money-related issues — Ahmed receives unexpectedly high electricity bills with no clear breakdown of what drove the cost.
Product	Quality issues — Smart meters and monitoring apps are unreliable, inaccurate, or difficult to set up and use.
Process	Related to the user's journey — Ahmed has no consistent routine to track usage; he relies on random manual checks.
Support	Getting help from customer service — Limited guidance from utility providers on how to interpret bills or reduce usage.

Note that:

- The problem may stem from difficulty using the technology itself, or from a negative prior experience with a similar product — *i.e., a confusing meter app vs. a simple, clean dashboard.*
- Everything that the product has to offer might seem obvious to the designer, but the design must reflect the mental model of the user, not the system.

3.2 5Ws and H Analysis

To create an accurate problem statement, the 5Ws and H framework was applied using insights drawn from the persona and empathy map. This analysis grounds the problem in Ahmed's real-world context:

Question	Answer
Who?	Ahmed Hassan — a 35-year-old homeowner in Cairo who wants real-time visibility into his household electricity consumption.
What?	Ahmed struggles to monitor and control his electricity usage in real time. He has no reliable tool to track which devices consume the most or to receive alerts when usage is unusually high.

Question	Answer
Where?	At home — primarily through a mobile app while managing daily household routines.
When?	The problem occurs continuously throughout the month, with peak frustration when the electricity bill arrives unexpectedly high.
Why?	Lack of real-time data causes Ahmed to waste energy unknowingly, leading to financial strain and anxiety about the bill.
How?	Ahmed currently checks his bill only on arrival, turns off devices randomly, and manually estimates usage — all without any data-driven insight.

| Stage 03: Ideate a Solution

Stage 03 marks the transition from the problem space to the solution space. Having defined Ahmed's pain points and problem statement, this stage develops a Goal Statement that sets the scope and direction for the Smart Meter Mobile Application design.

4.1 Goal Statement Approach

To focus the scope of our design, we create a Goal Statement: one or two sentences that describe the product and its benefits for the user. A well-formed goal statement answers three key questions:

Who	Who will the product serve? — Ahmed: a tech-savvy homeowner seeking real-time energy visibility.
What	What will the product do? — Provide real-time monitoring, usage dashboards, intelligent alerts, and energy-saving recommendations.
Why	Why does the product solve the user's need? — It eliminates the feedback delay between usage and awareness, empowering Ahmed to make informed, timely decisions about his energy consumption.

4.2 Goal Statement

Goal Statement

Our Smart Meter Mobile Application **will let users** monitor their electricity consumption in real time, receive intelligent alerts, and access personalized energy-saving recommendations, **which will affect** homeowners like Ahmed who currently have no reliable way to track or manage their energy usage throughout the month, **by** reducing bill shock, eliminating energy waste, and building confidence in managing household electricity efficiently. **We will measure effectiveness by** tracking reductions in average monthly electricity consumption, user-reported bill savings, frequency of alert engagement, and overall user satisfaction scores.

4.3 Proposed Application Features

Based on the goal statement and Ahmed's identified needs, the following five core features are proposed for the Smart Meter Mobile Application:

1. **Real-Time Energy Monitoring —** Displays live electricity consumption data collected directly from the smart meter, giving Ahmed instant visibility into current household usage.
2. **Usage Dashboard —** Visualizes daily, weekly, and monthly consumption trends through intuitive charts, enabling Ahmed to spot patterns and compare periods at a glance.
3. **Alerts & Notifications —** Automatically notifies Ahmed when consumption exceeds a threshold or when an unusual pattern is detected — addressing his fear of surprise bills.
4. **Energy Saving Recommendations —** Delivers personalized, data-driven suggestions to help Ahmed reduce usage and identify inefficient appliances.
5. **Usage History —** Provides a complete historical log of consumption data, allowing Ahmed to track progress, review past periods, and understand his long-term energy habits.

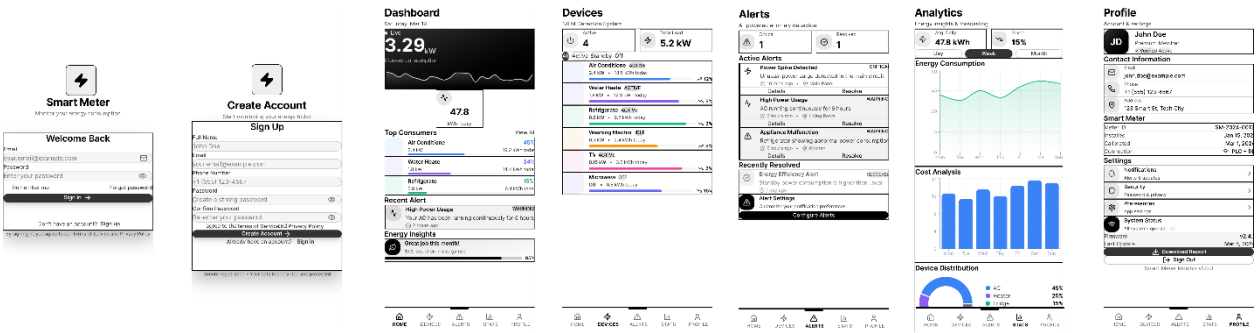
3.3 Problem Statement

Problem Statement

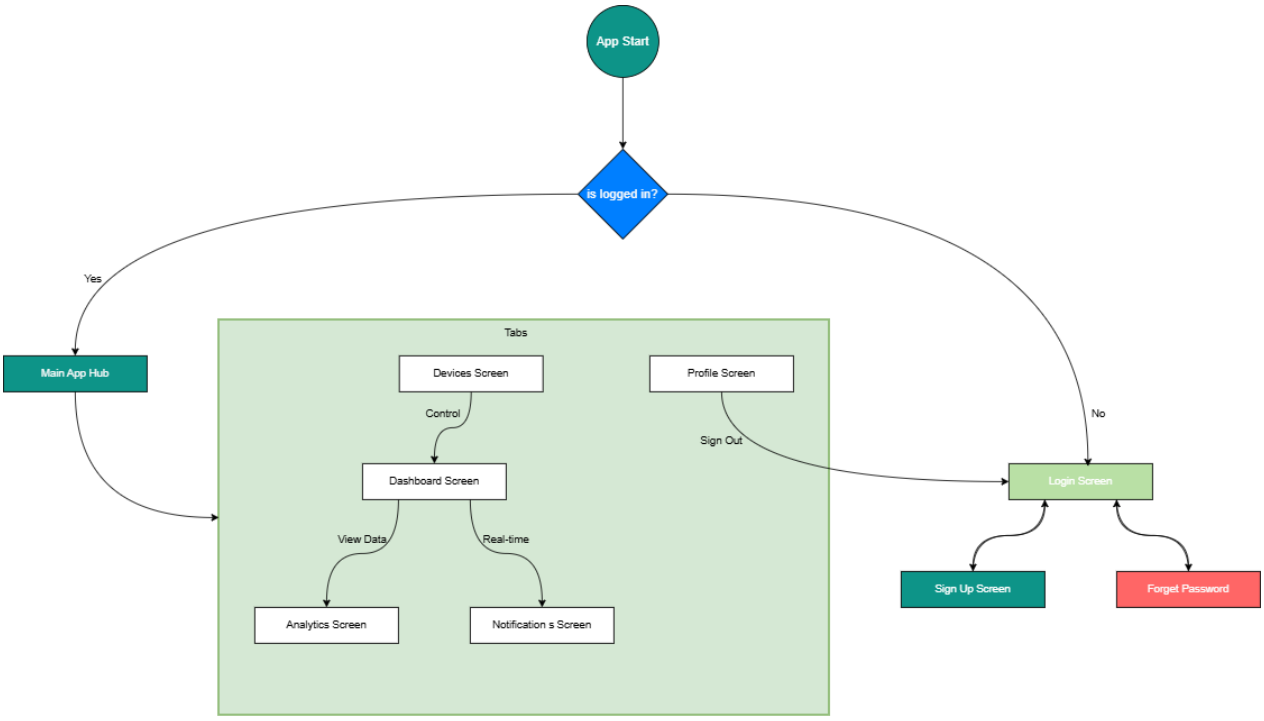
Ahmed is **a/an** tech-savvy homeowner who manages household energy manually **who needs** a reliable, real-time energy monitoring mobile application **because** the lack of visibility into his electricity consumption leads to unexpectedly high bills, energy waste, and anxiety — with no easy way to identify inefficient appliances or receive timely alerts about abnormal usage.

| Stage 04: WireFrame & UI\UX

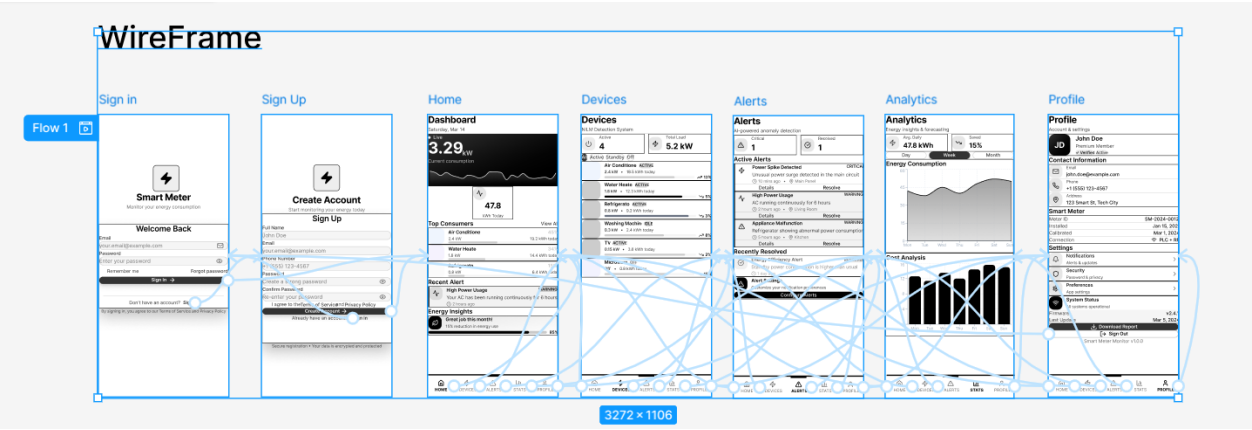
Stage 03 marks the transition from the problem space to the solution space. Having defined Ahmed's pain points and problem statement, this stage develops a Goal Statement that sets the scope and direction for the Smart Meter Mobile Application design.



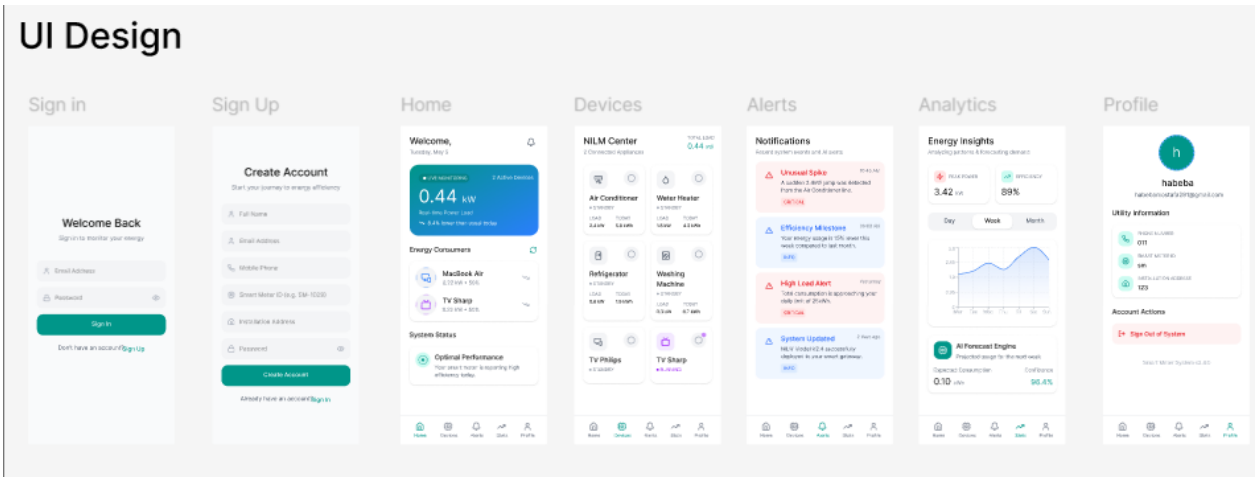
Site Map



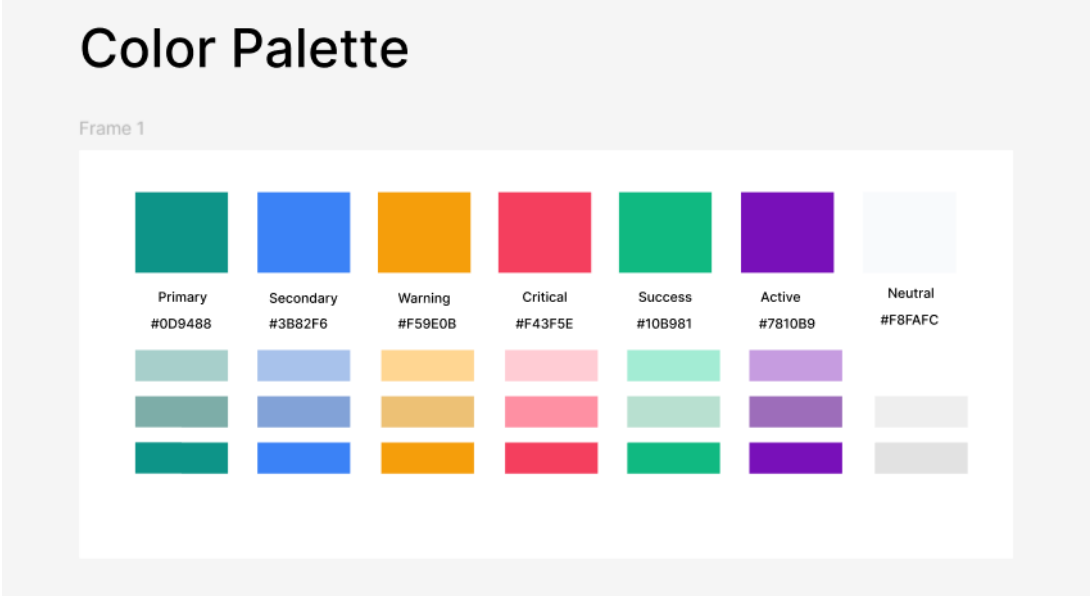
Wirframe and it is interactions prototype



UI Design



Section 02: Frontend



Typography (Font Hierarchy)

Role	Font Name	Size	Usage
Title	Inter Bold	32 - 48px	App name, Main headers, Large power values
Header	Inter SemiBold	20 - 24px	Screen titles, Section headers, Card titles
Regular	Inter Regular	14 - 16px	Body text, descriptions, form labels
Bold	Inter Bold	14 - 18px	Button labels, high-load values, statistics

Custom Buttons

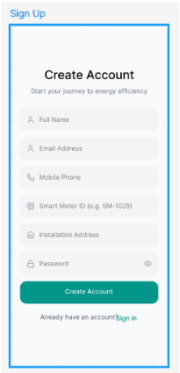
Component Name	Style	Where is used?	Component Design Brief
PrimaryButton	Solid Teal, Rounded	Login, Sign Up, Save	High-contrast, 16px radius, full-width design.
SecondaryButton	Outlined Indigo	Logout, Cancel, Reset	Transparent background, 2px border, clean look.
AuthToggle	Text Link	Login <-> SignUp	Subtle typography link with Primary color accent.

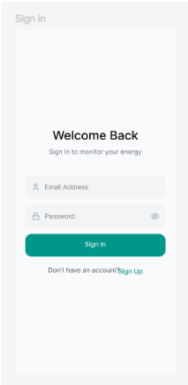
Other Custom Components

Component Name	Style	Where is used?	Component Design Brief
EnergyCard	Glassmorphism	Dashboard, Devices	Semi-transparent background with soft inner shadow.
AuthForm	Unified Container	Auth Flow	Single component that switches between Login/SignUp.
AlertBanner	Floating Toast	Dashboard (Top)	Critical rose background, slides from top on Anomaly.
PulseIcon	Animated SVG	Dashboard	Teal circle with a "breathing" animation for live data.

Screenshots for UI Design

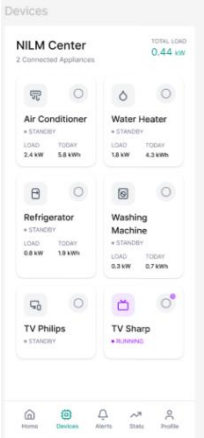
1. The Unified Auth Gateway (Login/Sign-Up)
Description: A minimalist, high-contrast entry point that uses a unified AuthForm component. It features a clean white background with Energy Teal accents. The design focuses on "Zero-Distraction" UX, using smooth transitions between Login and Registration modes to reduce user friction.



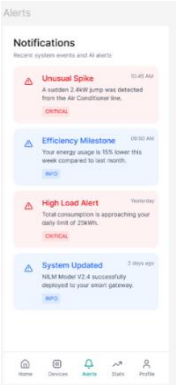


2. The Live Intelligence Dashboard
Description: The heart of the application. It utilizes a Glassmorphic Hero Card that displays a real-time "Consumption Wave." A breathing pulse animation indicates active AI monitoring. The layout uses a 2-column grid for quick stats (Daily Cost, Voltage, Current), ensuring the most critical data is visible at a glance.

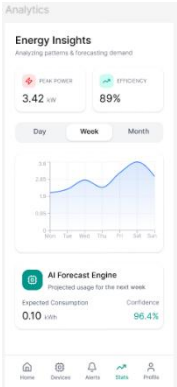
3. The Appliance Control Hub (Devices)
Description: A vibrant, card-based interface showing individual appliance statuses. Each card features a Staggered Entrance Animation and a "Glow State"—a soft color aura (e.g., Emerald for Fridge, Amber for AC) that appears only when the AI detects the device is active.



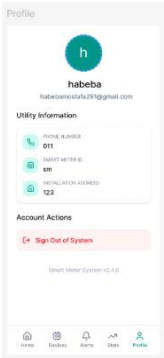
4. The Deep-Insight Analytics
Description: A sophisticated Dark-Mode analytical suite. It uses a deep Indigo-to-Black gradient to make the energy forecasting charts "pop." The screen includes an AI Confidence badge and a dynamic wave chart that visualizes predicted consumption patterns for the upcoming week.



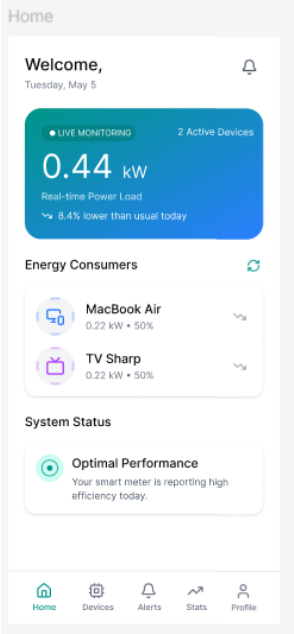
5. The Smart Notification Center (Alerts)
Description: An elegant, button-free notification log. Alerts are categorized by color: Critical Rose for energy spikes and Data Indigo for efficiency updates. The design uses semi-transparent "Glass" rows with 24px corner radii, following the modern mobile "Toast" notification aesthetic.

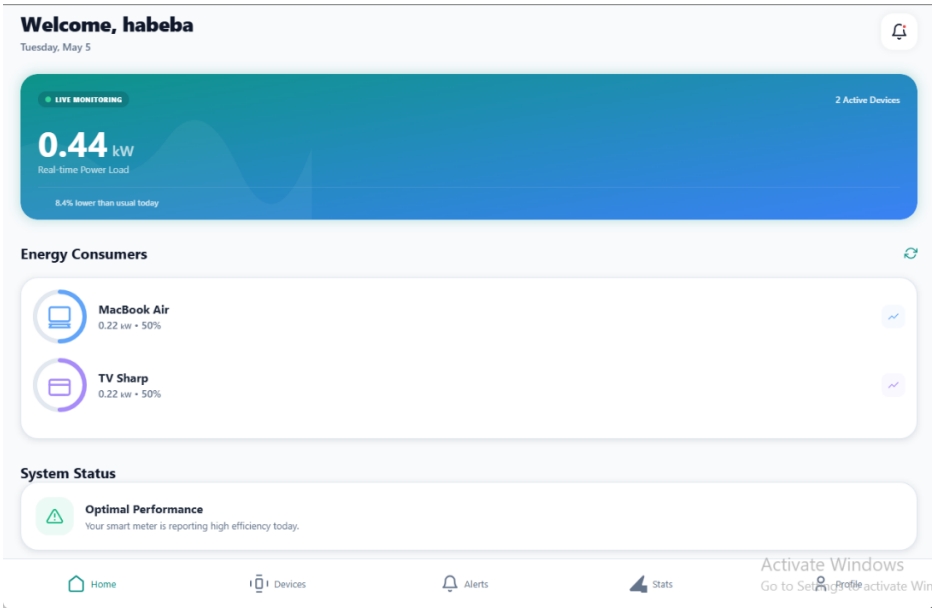


6. The Utility Command Center (Profile)
Description: A personalized space for the user. It features a Glowing Avatar Ring and a structured "Information Grid" that fetches the User's Phone, Meter ID, and Home Address from Firestore. The layout is designed to feel like a giving the user a sense of ownership over their energy data.



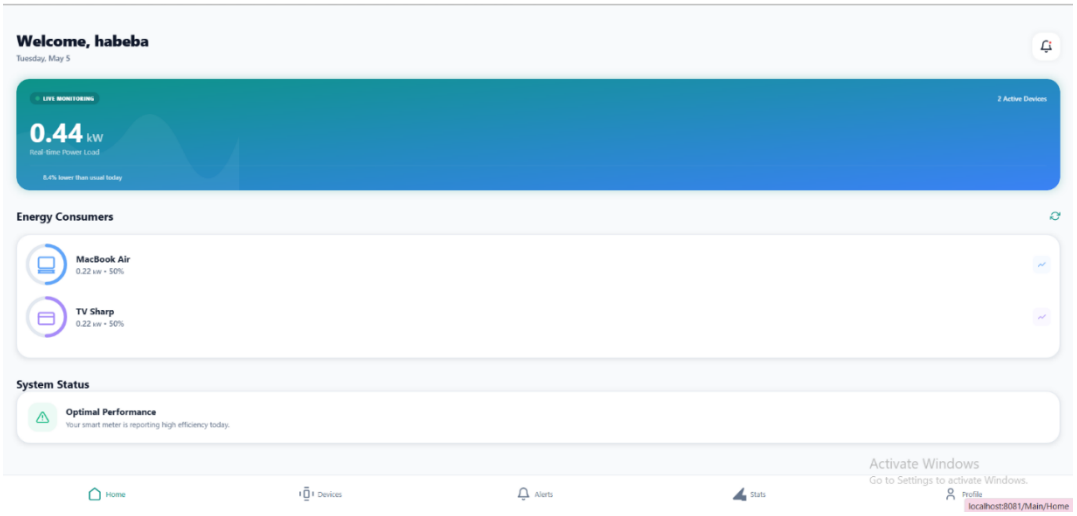
Responsiveness





Phone

Tablet



Laptop

Section 03: Backend & Dynamic Architecture

1. Core Infrastructure (Firebase & Firestore)
The backbone of the Smart Meter Monitoring System is built on **Google Firebase**. We utilized a NoSQL **Cloud Firestore** database to handle real-time data streaming. Unlike traditional databases, Firestore allows the app to listen for changes and update the UI instantly without the user needing to refresh the page.

- **Authentication: Managed by Firebase Auth, providing secure, encrypted entry for users.**
- **Real-time Streaming: Data is stored in collections like data_nilm (for sensor readings) and users (for profile metadata).**

2. The Dynamic Content Pipeline
Your app's content is not static; it "breathes" with the incoming energy data. Here is the technical flow of how data is rendered:

1. **Data Ingestion**: A Python script or a physical IoT gateway pushes raw electrical parameters (Voltage, Current, Active Power) into the Firestore data_nilm collection every few seconds.

2. **State Synchronization**: Inside the React Native app, we use React Hooks **(useEffect and useState)** combined with Firestore queries to fetch the most recent 400 entries of data.

3. **AI Processing (The Intelligence Layer)**:
    - Once the app retrieves the dynamic data from the database, it doesn't just show numbers.
    - It pipes this raw data into our Hugging Face AI Models.
    - The backend processes this via an API call and returns a "Disaggregated Load" (NILM) and "Anomaly Score."

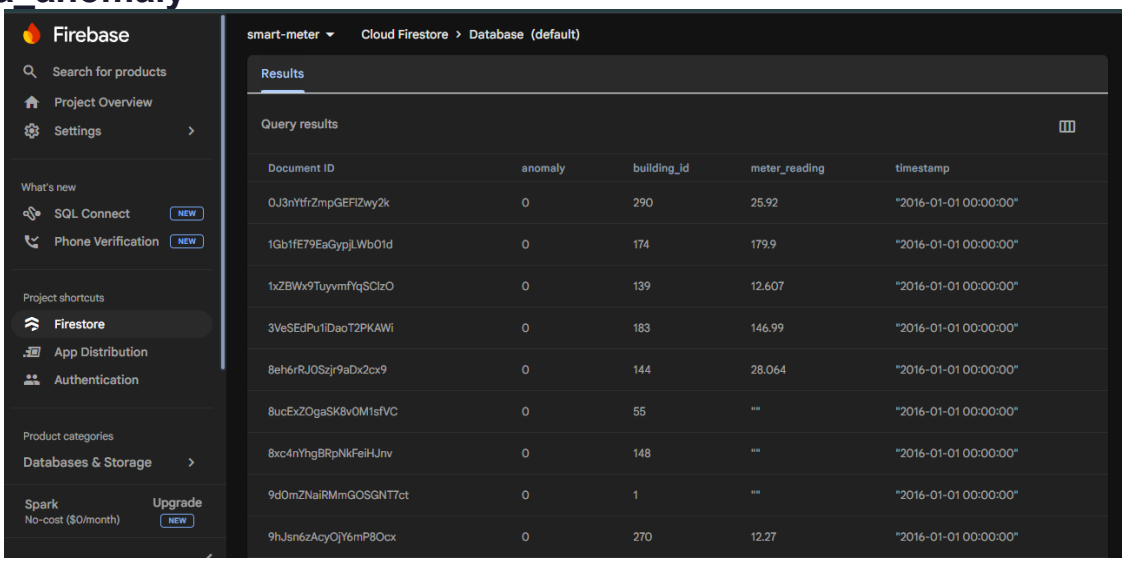
4. **Dynamic UI Rendering**:
    - The **Dashboard** renders a "Live Wave" and "Power Value" directly from the database.
    - The **Devices Screen** dynamically updates the "ON/OFF" status of appliances based on the AI's real-time interpretation of the database records.
    - The **Profile Screen** fetches the user's specific Installation Address and Meter ID from the users document keyed to their Unique ID (UID).

3. Backend Technology Stack Table

Technology	Role	Description
Cloud Firestore	Primary Database	Real-time NoSQL storage for energy logs and user metadata.
Firebase Auth	Security Layer	Handles encrypted Login, Sign-Up, and session management.
Hugging Face API	AI Inference	Hosts our NILM, Anomaly, and Forecast models on a FastAPI server.
Python (Backend)	Data Simulation	Used for cleaning and uploading raw CSV datasets into the live cloud stream.

Screens of the Backend

data_anomaly



Document ID	anomaly	building_id	meter_reading	timestamp
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1Gb1fE79EaGypjLWb01d	0	174	179.9	"2016-01-01 00:00:00"
1xZBWx9TuyvmfYqSClzO	0	139	12.607	"2016-01-01 00:00:00"
3VeSEdPu1iDaoT2PKAWi	0	183	146.99	"2016-01-01 00:00:00"
8eh6rRJ0Szjr9aDx2cx9	0	144	28.064	"2016-01-01 00:00:00"
8ucExZOgaSK8v0M1sfVC	0	55	""	"2016-01-01 00:00:00"
8xc4nYhgBRpNkFeiHJnv	0	148	""	"2016-01-01 00:00:00"
9d0mZNaiRMmGOSGNT7ct	0	1	""	"2016-01-01 00:00:00"
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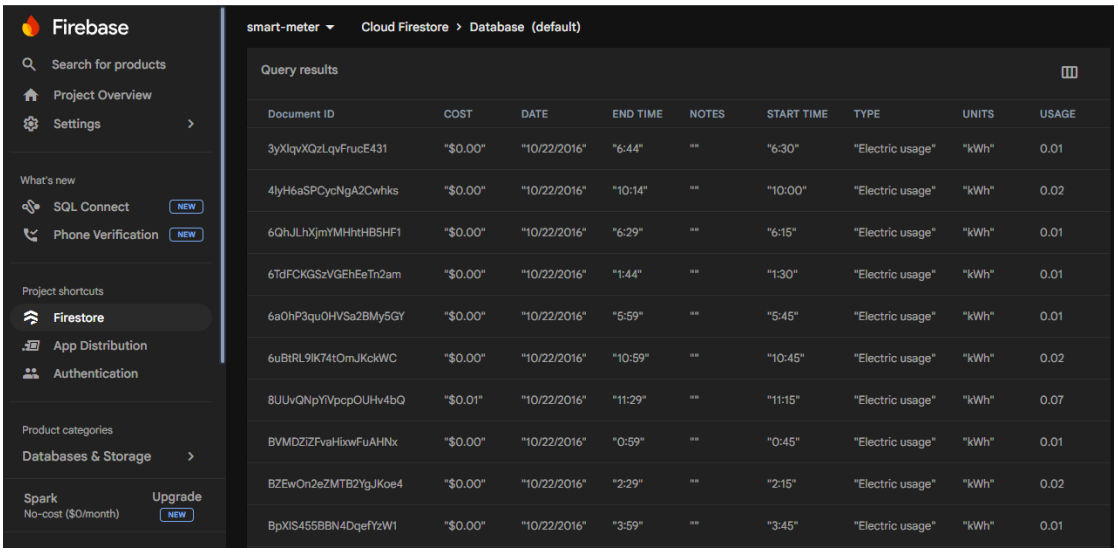
data_forecast



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4IyH6aSPCycNgA2Cwhks	"$0.00"	"10/22/2016"	"10:14"	""	"10:00"	"Electric usage"	"kWh"	0.02
6QhJLhXjmYMHhtHB5HF1	"$0.00"	"10/22/2016"	"6:29"	""	"6:15"	"Electric usage"	"kWh"	0.01
6TdFCKGSzVGEhEeTn2am	"$0.00"	"10/22/2016"	"1:44"	""	"1:30"	"Electric usage"	"kWh"	0.01
6a0hP3qu0HVSa2BMy5GY	"$0.00"	"10/22/2016"	"5:59"	""	"5:45"	"Electric usage"	"kWh"	0.01
6uBtRL9lK74tOmJKckWC	"$0.00"	"10/22/2016"	"10:59"	""	"10:45"	"Electric usage"	"kWh"	0.02
8UUvQNpYiVpcpOUHv4bQ	"$0.01"	"10/22/2016"	"11:29"	""	"11:15"	"Electric usage"	"kWh"	0.07
BVMDZiZFvaHixwFuAHNx	"$0.00"	"10/22/2016"	"0:59"	""	"0:45"	"Electric usage"	"kWh"	0.01
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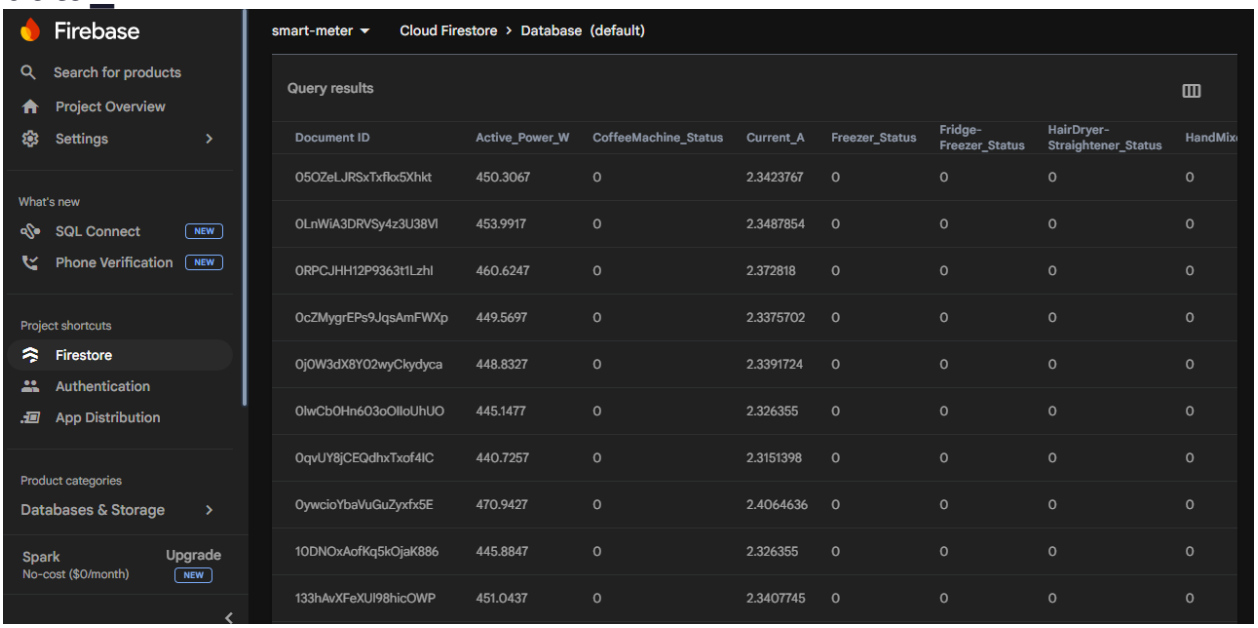
data_nilm



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ORPCJHH12P9363t1LzhI	460.6247	0	2.372818	0	0	0	0
OcZMygrEPs9JqsAmFWXp	449.5697	0	2.3375702	0	0	0	0
Oj0W3dX8Y02wyCkydyca	448.8327	0	2.3391724	0	0	0	0
OlwCb0Hn6O3oOIIoUhUO	445.1477	0	2.326355	0	0	0	0
OqvUY8jCEQdhxTxof4IC	440.7257	0	2.3151398	0	0	0	0
OywcioYbaVuGuZyxfx5E	470.9427	0	2.4064636	0	0	0	0
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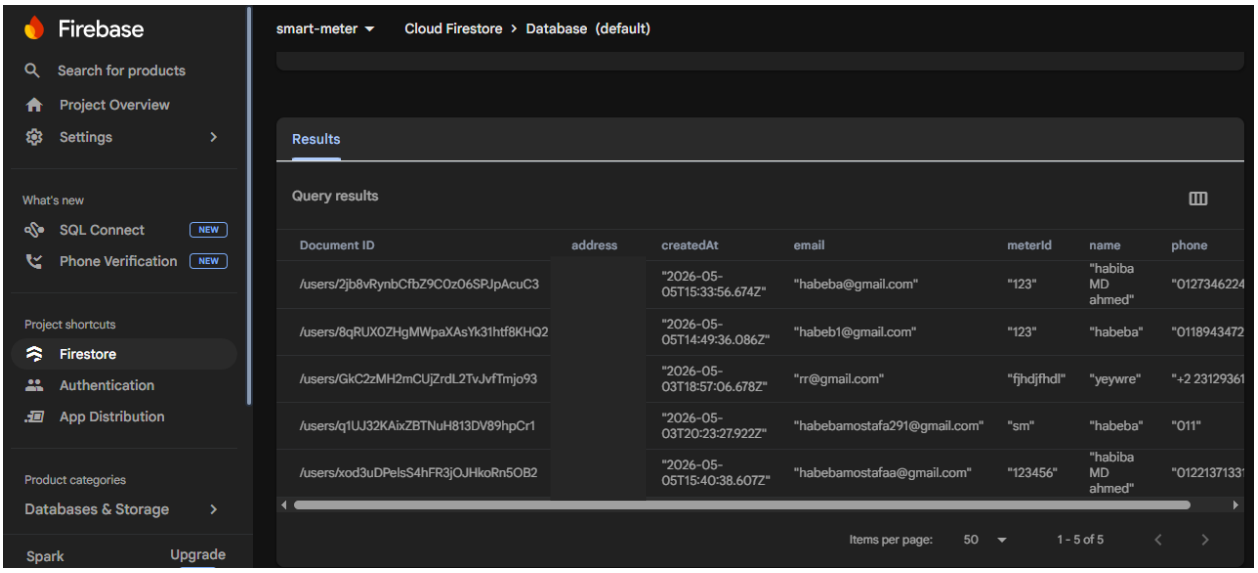
users



Document ID	address	createdAt	email	meterId	name	phone
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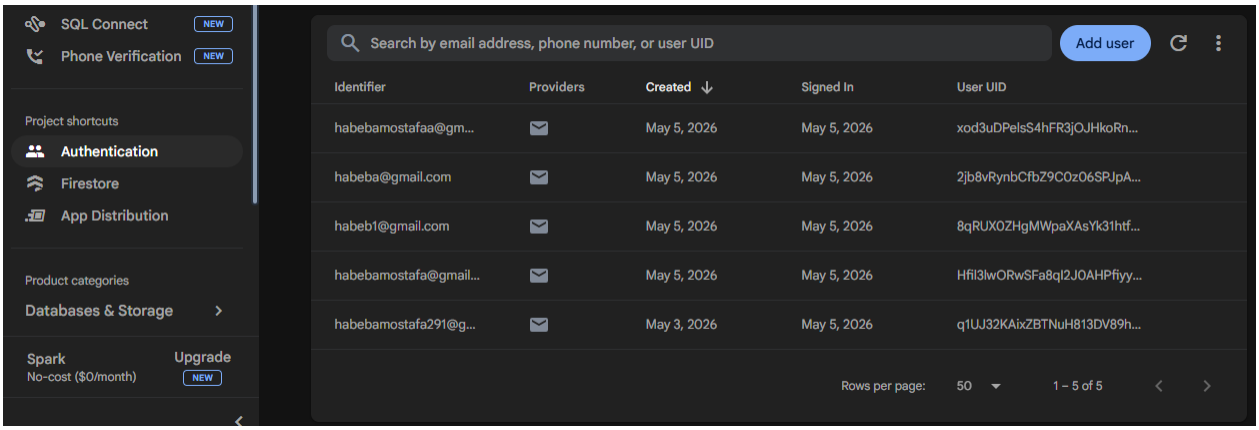
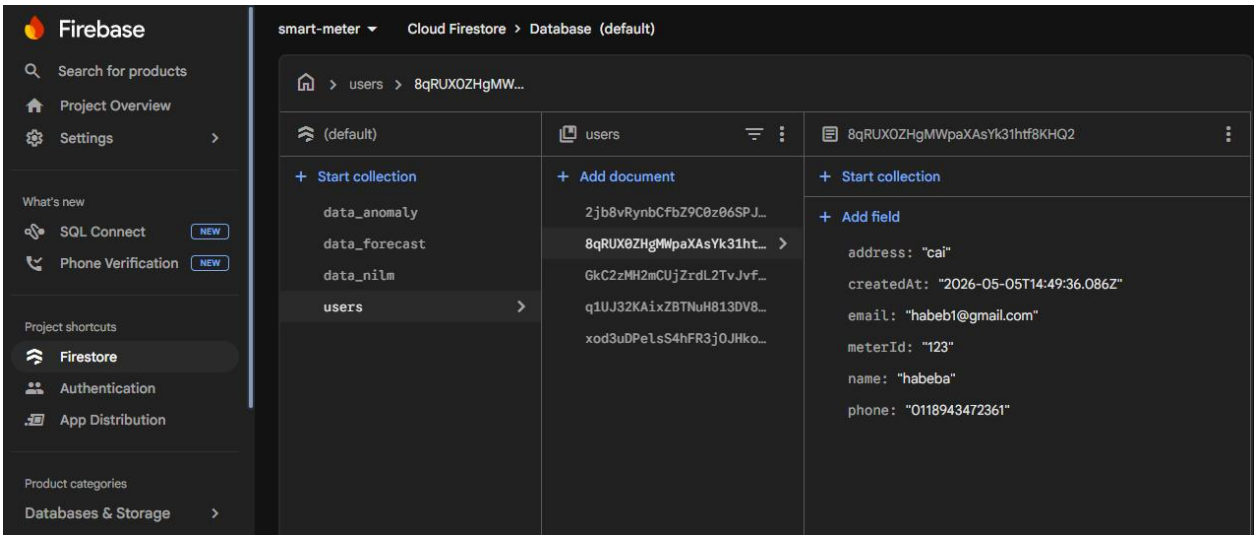
Authentication



general overview